

Weekly Report: Ultra-High-Pressure ODMR — Why should the anvil culet be [111]? Testing the $E = 0$ argument

Risei Abe

Week: 2026/08/02 – 2026/08/08

Question

A [111] culet puts one NV family exactly along the load axis, so that family keeps its C_{3v} symmetry and its transverse-strain splitting E vanishes; does that argument actually make [111] the better choice for ODMR sensitivity at 120 GPa, compared with a [100] culet where all four families are equivalent?

Hypothesis

If the axial family on a [111] culet has $E = 0$ and therefore no transverse-strain broadening, it should give the narrowest line and win despite carrying only 1/4 of the NV population, whereas [100] keeps all four families but pays a linewidth penalty $\propto |dE/dt| \delta t$ from the spread δt in deviatoric stress across the culet.

Evidence / Interpretation

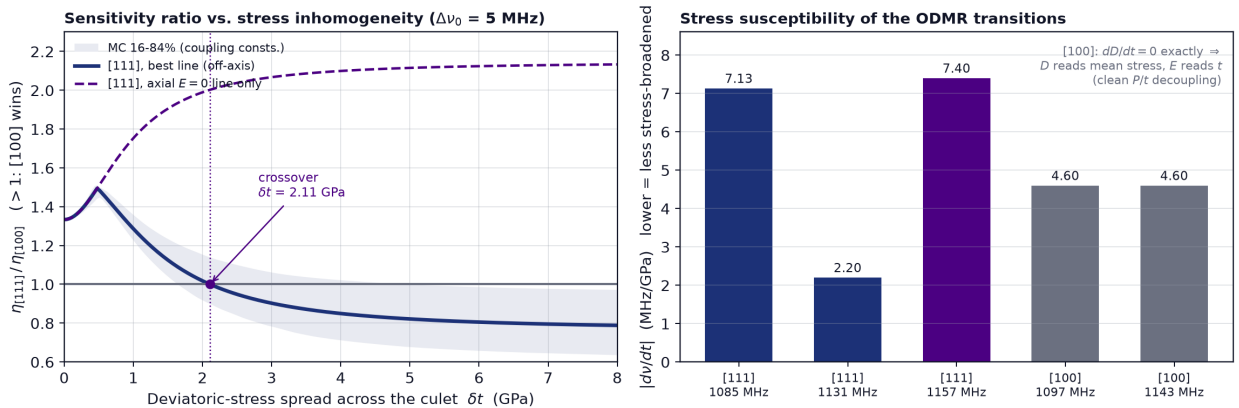


Fig 1: Left: $\eta_{[111]}/\eta_{[100]}$ vs. the deviatoric-stress spread δt , for the best [111] line and for the axial $E = 0$ line alone, with a Monte-Carlo band over the Barson spin-stress constants. Right: $|d\nu/dt|$ of every line; the $E = 0$ line is the steepest one.

- The symmetry argument itself holds exactly ($E = 0$ and $dE/dt = 0$ for the axial family),

but that line is nevertheless the *most* deviatoric-stress-sensitive one in the problem, $|dD/dt| = |2a_2| = 7.40 \text{ MHz/GPa}$ versus 4.60 MHz/GPa for $[100]$, and with only 0.375 of the PL dipping it loses to $[100]$ by $1.33\times$ to $2.14\times$ at every δt .

- $[111]$ does win, but above $\delta t \approx 0.42 \Delta\nu_0$ (GPa per MHz of intrinsic width, i.e. 2.1 GPa for $\Delta\nu_0 = 5 \text{ MHz}$) and through the *off-axis* families, whose line at $|d\nu/dt| = 2.20 \text{ MHz/GPa}$ is the narrowest available — so the correct reason to pick $[111]$ is the opposite of the usual one.

Gap / Failure

- The model scores only $\eta \propto \Delta\nu/(C\sqrt{R})$ at zero bias field, so it cannot see the genuine $[111]$ advantage for vector magnetometry, where a field along the axial NV is unmixed by E .
- It also says nothing about the dominant real-world reason for $[111]$ anvils — the higher compressive strength and the cleavage-plane geometry that decide whether the anvil survives 120 GPa at all.

Next Step

Fold the δt -dependent linewidths from this model into the background-aware sensitivity code so that the culet choice, the excitation wavelength and the background level are optimised together rather than one at a time.

References

1. M. S. J. Barson *et al.*, “Nanomechanical Sensing Using Spins in Diamond,” *Nano Lett.* **17**, 1496 (2017) — spin-stress constants a_1, a_2, b, c .
2. P. Udvarhelyi *et al.*, *Phys. Rev. B* **98**, 075201 (2018) — spin-strain Hamiltonian conventions.
3. M. W. Doherty *et al.*, *Phys. Rev. Lett.* **112**, 047601 (2014) — NV under pressure.
4. Repository: `code/anvil_orientation.py`, `code/fig6_anvil_orientation.py` (this analysis).